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BEVERLY INDEX LLC · SHERIDAN, WYOMING
NOUS Platform · Scientific Governing Document

Scientific Principles & Governing Architecture

The non-negotiable scientific constraints, architectural decisions, and theoretical foundations that govern every component of the NOUS Platform. All code, all models, all outputs must be consistent with this document. Conflicts between implementation convenience and these principles are resolved in favor of these principles — always.

V1.1 · MARCH 2026 | GOVERNING DOCUMENT | ACTIVE

AUTHORSHIP & AUTHORITY

This document was authored by Randolph R Beverly Jr, President & Chief Architect, Beverly Index LLC, and reflects the accumulated scientific framework, architectural decisions, and design principles established across the full NOUS Platform development history. It supersedes any implementation detail that contradicts it. If code behaves contrary to a principle stated here, the code is wrong.

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01 *The Core Detection Chain*

NON-NEGOTIABLE

The governing logic of the entire NOUS Platform is a three-link detection chain. Every component, every model, every output must be understood as an expression of this chain. The chain must never be broken, bypassed, or short-circuited.

LINK 1
Neural Oscillatory State
(invisible, internal)

↓

LINK 2
Cognitive Processing Configuration

↓

LINK 3
Linguistic Output Features

↓

NOUS OUTPUT
NOSS Activation Vector

P-01.1

NOUS detects cognitive consequence, not pathological cause

NOUS does not diagnose blood, tissue, or genetic conditions directly. It detects the downstream cognitive-linguistic consequences of those conditions. This is not a limitation — it is the scientific claim. The system provides: "this oscillatory pattern is consistent with X." Medical investigation identifies the cause.

VIOLATION EXAMPLE →

```
Never claim NOUS "detects a tumor," "detects a clot," or "detects a gene mutation."
Correct claim: "NOUS detected cognitive signatures consistent with chemotherapy
neurotoxicity." The cause is for clinicians to determine.
```

P-01.2

Indirect measurement through secondary effects is the method

The primary phenomenon (neural oscillatory states) is inaccessible during natural conditions. NOUS detects secondary effects — the cascade of linguistic and prosodic features that oscillatory states produce when they encounter natural language production. This is analogous to how the

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Telescope Array reconstructs cosmic ray energy from secondary particle cascades in the atmosphere: the primary event is never directly observed; the downstream effects are measured with precision.

CODE IMPLICATION →

```
All model training, feature extraction, and output calibration must be designed around
measurable linguistic and prosodic secondary effects. No claim of direct neural
measurement should ever be made in code comments, documentation, or outputs.
```

02 *NOST Theoretical Foundations*

SCIENTIFIC

Neural Oscillation Signature Theory (NOST) holds that distinct neural oscillatory states produce distinct, computationally extractable signatures in communicative output. The theory rests on five formal propositions, each of which must be honored in implementation.

P-02.1 Oscillatory Governance — Proposition 1

Cognitive and emotional states are governed by neural oscillatory dynamics. The state of a person's cognitive architecture at any moment is reflected in the frequency-domain properties of neural activity across multiple brain networks. Delta (1–4 Hz), Theta (4–8 Hz), Alpha (8–12 Hz), Beta (13–30 Hz), and Gamma (30–100 Hz) bands each carry distinct functional significance.

P-02.2 Linguistic Encoding — Proposition 2

Neural oscillatory states encode into linguistic output. The way a person constructs narratives, selects vocabulary, structures syntax, and produces prosody are downstream expressions of their neural oscillatory configuration. Language is not merely communication — it is a biomarker.

CODE IMPLICATION →

```
REMIEL's 8-channel feature extraction is the computational operationalization of this
proposition. Each channel targets a specific class of linguistic features that corresponds
to specific oscillatory band activity.
```

P-02.3 Signature Specificity — Proposition 3

Different clinical conditions produce distinguishable oscillatory signatures that persist even under extreme conditions. A person in acute psychosis produces a different NOSS vector than a person with generalized anxiety — not just quantitatively but qualitatively. These signatures are resilient: they survive deception attempts, psychological defense mechanisms, and deliberate concealment. The subject cannot consciously suppress their oscillatory signature.

Supporting physics analogy (for publications only): CERN quarkonium resonance survival findings

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(2025) demonstrate that bound-state resonances persist as detectable signatures within quark-gluon plasma at temperatures far beyond their expected dissolution point — providing cross-domain precedent for structured-state resilience under extreme conditions.

CODE IMPLICATION →

The CSAA pathway is built on this proposition. Fabricated narratives cannot fully suppress the oscillatory signature of the cognitive process generating them. The authenticity analysis measures the ratio between perceptual-dominant and executive-dominant encoding patterns.

P-02.4

Clinical Correspondence — Proposition 4

NOSS sector activations correspond to clinically validated diagnostic categories defined in DSM-5/ICD-11. This correspondence is empirical, not definitional — it is to be validated by clinical studies, not assumed by architecture. The 12-sector NOSS classification was designed with DSM-5/ICD-11 alignment as a primary design constraint.

P-02.5

Treatment Sensitivity — Proposition 5

Effective treatment produces measurable changes in oscillatory signature over time. This is the foundation of NOUS's longitudinal monitoring value: if treatment works, NOSS activation patterns should shift measurably. If they don't, the treatment may not be working — regardless of patient self-report. This proposition makes NOUS a treatment response monitoring tool, not just a diagnostic instrument.

CODE IMPLICATION →

VOCA dual-rate baseline learning (fast EMA α=0.1, slow EMA α=0.01) is the operational implementation of this proposition for continuous monitoring. The trajectory engine tracks sector-level changes across sessions to measure treatment response.

P-02.6

Lense-Thirring Temporal Principle — Assessment Timing

The most diagnostically informative signatures emerge during the post-acute consolidation period, not during peak crisis. Analogous to deuteron formation in nuclear physics — fragile structures form during cooling, not during the collision. A narrative collected 48 hours post-event may produce a more reliable NOSS signature than one collected during acute distress. AAE protocol design must account for this.

CODE IMPLICATION →

AAE should track whether a session is being conducted during acute crisis and apply appropriate confidence modifiers. The Confidence Impact statement in the AAE configuration card must reflect assessment timing as a reliability factor.

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03 *NOSS — The 12-Sector Activation System*

NON-NEGOTIABLE

The Neural Oscillatory Spectroscopy System (NOSS) is a 12-dimensional continuous activation vector. It is not a categorical classifier. Each sector produces a continuous value in the range [0, 100]. The full output is:

```
NOSS_vector = a ∈ [0, 100]¹²
// Where each aᵢ is a continuous real-valued activation, NOT a discrete category
// Multi-sector co-activation is expected and clinically meaningful
// No single sector exclusively "owns" any diagnostic condition
```

SECTOR	DOMAIN	PRIMARY CLINICAL CORRESPONDENCE	NOTES
S1	Mood	MDD, Bipolar (depressive phase)	
S2	Anxiety	GAD, Panic Disorder, PTSD (hyperarousal)	
S3	Psychotic	Schizophrenia, Schizoaffective, Acute Psychosis	
S4	Neurocognitive	Executive dysfunction, Processing speed deficit, TBI	
S5	Developmental	ASD, ADHD, Intellectual disability, Down Syndrome	
S6 INVERSE	Optimal	Cognitive wellbeing, Integration, Resilience	Structurally inverse to distress. MUST be dampened when S1/S2/S3 elevated.
S7	TBI	Traumatic Brain Injury, Concussion, CTE	
S8	Neurodegenerative	Parkinson's (prodromal), Alzheimer's, MCI	4.7 Hz focal tremor is prodromal Parkinson's marker
S9	Captivity/Trauma	Complex PTSD, Captivity trauma, Torture	
S10	Substance	Substance use disorder, Dependency patterns	
S11	Dissociative	DID, Depersonalization, Dissociative amnesia	
S12	Personality	Cluster B personality disorders, BPD	

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HARD RULE — S6 INVERSION

When S1, S2, or S3 exceed 50, S6 activation must be dampened proportionally. A person in psychotic crisis (S3=100) cannot simultaneously show high cognitive wellbeing (S6=80). The current NOSS scoring engine must enforce this constraint. Absence of this constraint is a calibration failure.

HARD RULE — CONTINUOUS VECTOR INTEGRITY

Downstream classifiers must consume the full 12-dimensional vector, not a single dominant sector. The Beverly Index DiagnosticProbabilityScorer must weight multiple sectors in its condition probability calculations. Mapping a single sector value to a diagnosis without considering the full vector is a classification error.

04 *Non-Negotiable Scoring Principles*

NON-NEGOTIABLE

P-04.1 Confidence Decouples from Severity

Confidence affects diagnostic specificity — how certain we are *which* condition is present. It does not reduce urgency or clinical action priority. A single session with S3=100 is CRITICAL even at 36% confidence. A CRITICAL finding with low confidence means: "We are highly confident something serious is happening; we are less certain of the specific diagnosis." Both signals matter. Neither cancels the other.

VIOLATION EXAMPLE →

```
ANTI-PATTERN: if confidence < 0.5: priority = "LOW" — This directly violates this
principle and represents a safety hazard.
```

CODE IMPLICATION →

```
ClinicalActionPriority must read NOSS sector activations directly, with sector thresholds
determining priority level independent of confidence score. Confidence modifies the
accompanying text, not the priority tier.
```

P-04.2 Beverly Index v2.0 — 7 Components, Priority-Led

The Beverly Index has exactly 7 components. ClinicalActionPriority is Component 1 and the headline metric for CI pathway assessments. The scoring hierarchy is:

```
Component 1: ClinicalActionPriority ← HEADLINE (CI pathway)
Component 2: DiagnosticProbability
Component 3: NarrativeCoherence
Component 4: IntegrationDynamics
Component 5: PatternClassification
Component 6: ConfidenceScore
```

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```
Component 7: AuthenticityIndex ← CSAA pathway ONLY
```

HARD RULE – AUTHENTICITYINDEX SCOPE

AuthenticityIndex (Component 7) is computed and displayed only when the CSAA pathway is active. It must never appear in CI or VOCA pathway outputs.

P-04.3 Pathway-Adaptive Headline Metric

The Beverly Index headline metric adapts based on active pathway:

```
CI pathway: Clinical Action Priority (0–100, higher = more urgent)
CSAA pathway: Forensic Confidence Rating (%, higher = more confident)
VOCA pathway: Deviation from Baseline (%, higher = more change detected)
```

All three use the same underlying computational engine with pathway-aware output framing. They are not separate scoring systems.

P-04.4 Psychotic Sector Priority Hierarchy

When S3 (Psychotic) exceeds 70, the top diagnostic condition must be Psychotic spectrum. No other sector can override S3's primacy at this activation level. Specifically:

```
IF S3 > 70 → Psychotic/Schizophrenia spectrum LEADS, regardless of other sectors
IF S3 > 40 AND S1 > 40 → Schizoaffective LEADS over pure Bipolar
Bipolar LEADS ONLY when: S1 > 60 AND S3 < 50
MDD LEADS ONLY when: S1 > 50 AND S3 < 30
```

VIOLATION EXAMPLE →

ACTIVE BUG (being fixed): S3=100 mapping to "Bipolar" as top condition. This is a direct violation of the hierarchy above.

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05 *Platform Architecture Hard Rules*

ARCHITECTURE

AAE The *only* user-facing entry point into the platform. AAE auto-routes to pathways based on input characteristics, with manual override capability for practitioners. Users never interact with REMIEL, NOSS, or Beverly Index directly. AAE is the orchestrator; everything HARD

else operates beneath it.

REMIEL	8-channel feature extraction engine. Channels run in parallel. The engine block inside the vehicle — never user-facing. Real NLP via spaCy, sentence-transformers, VADER, NRCLex, and librosa. All eight channels must be active for a full assessment; partial channel results carry explicit confidence penalties.	ARCH
LOGOS	Cross-cultural calibration layer. Executes before REMIEL on every assessment. Builds an individual Linguistic-Cultural Frame from the subject's own baseline — NOT a fixed population offset. Italian expressiveness_baseline=75 is a default; an individual's measured baseline overrides the population default.	HARD
R2X	Translation bridge. Maps REMIEL feature vectors directly to NOSS sector activations. The R2X mapping must be direct — REMIEL feature → NOSS sector — without an intermediate band-power abstraction layer that loses signal specificity. The R2X calibration fix (commit 576be0b) resolved an inversion caused by intermediate indirection.	HARD
RTCA	Real-Time Call Analysis. Classification: DEPLOYMENT PATHWAY, not an intelligence module. RTCA feeds data into the intelligence layer as input; it does not belong in the intelligence modules list. The intelligence layer has exactly three modules: THANATOS, ORACLE, AEGIS.	HARD
VOCA	Dual-rate EMA baselines: fast α=0.1, slow α=0.01. Five-session establishment period before VOCA-VERIFIED status. Four alert tiers: GREEN/YELLOW/ORANGE/RED. Cannot establish VOCA-VERIFIED baseline while subject is under active central nervous system depressant use — system must flag as VOCA-CONDITIONAL.	ARCH
CSAA	Cognitive Signature Authenticity Assessment. Authenticity analysis rests on the encoding ratio: authentic recall is perceptual-dominant, sensory-rich, temporally non-linear. Fabricated narrative is executive-dominant, structurally controlled, perceptually impoverished. The Authenticity Index must implement this at the feature extraction level, not as a post-hoc scoring adjustment.	ARCH
Beverly Index	Universal output layer. Activated by every pathway. The inputs to the 7 components differ by pathway; the scoring engine itself is shared. Treatment Intelligence populates below the Beverly Index in clinical reports: 4 local matches + 4 worldwide matches, evidence-leveled, NOSS-sector-aligned.	ARCH

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06 *The Four Deployment Pathways*

ARCHITECTURE

CI

Cognitive Intelligence
Clinical Assessment

Headline Metric

Clinical Action Priority

Primary Users

Clinicians, therapists, health systems, sports medicine

CSAA

Cognitive Signature Authenticity Assessment
Forensic Pathway

Headline Metric

Forensic Confidence Rating (%)

Primary Users

Intelligence analysts, forensic investigators, legal proceedings, insurance SIU

VOCA

Vocal Oscillation Cognitive Archive
Continuous Monitoring

Headline Metric

Deviation from Baseline (%)

Primary Users

Longitudinal clinical monitoring, telehealth, sports team tracking, occupational health

RTCA

Real-Time Call Analysis
Streaming Deployment Adapter

Function

Sentinel / Streaming input to CI or CSAA

Threshold

Triggers AAE structured assessment at activation threshold 65. 30s windows, 15s overlap.

HARD RULE — MULTI-PATHWAY ASSESSMENTS

All four pathways can run simultaneously on the same input in a Comprehensive Assessment. AAE runs the pipeline once through REMIEL; R2X generates outputs for all active pathways simultaneously. The Beverly Index report includes all active pathway results in a unified document. This is the "Comprehensive Cognitive Assessment" premium product tier.

07 *Three Intelligence Modules*

NON-NEGOTIABLE

There are exactly three intelligence modules. RTCA is not an intelligence module. Any documentation, code, or output listing four intelligence modules is incorrect.

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THANATOS	Mortality forecasting. NOSS delta decay tracking. MCI weights: S4=1.5, S6=1.5, S8=1.4, S7=1.3. TLMP 5 markers. Horizons: 6/12/36 months. Five risk tiers. Inputs from longitudinal NOSS trajectory data.	MODULE 1
ORACLE	Pharmacological response prediction. PRI (Pharmacological Response Index), ARS (4 flags), TWE (7 drug classes), TTM 12-week sigmoid curves. Predicts treatment response before prescription based on NOSS profile and existing baseline data.	MODULE 2
AEGIS	Cognitive fitness certification. CPS formula: S6×40% + S4×30% + complexity×15% + speech×15%. Three certification tiers: CERTIFIED (8h validity, threshold ≥80), PROVISIONAL (threshold 60–79), DENIED (threshold <60). FFD (Fitness for Duty), IDA, LPB Welford algorithm.	MODULE 3

08 *LOGOS Calibration Principles*

NON-NEGOTIABLE

P-08.1

LOGOS calibrates against the individual's own baseline — not population norms

LOGOS does not apply a fixed cultural correction factor. It builds an individual Linguistic-Cultural Frame (LCF) from the subject's own prior assessments and their cultural-linguistic profile. The Italian expressiveness_baseline=75 is a population default used only when no individual baseline exists. Once an individual baseline is established, it supersedes the population default.

VIOLATION EXAMPLE →

```
WRONG: apply Italian_correction_factor = 0.57 (fixed offset) CORRECT: calibrate against subject's own established LCF; use population default only for first-session subjects with no prior data
```

CODE IMPLICATION →

```
A therapist reviewing an Anxiety score of 31% must know it was 54% pre-calibration. The LOGOS calibration delta must always be visible in the AAE Configuration Card. Hiding this delta from the clinician is an information integrity failure.
```

P-08.2

Physiological and pharmacological conditions modify the LCF, not the NOSS output

When a subject has a cold, is on painkillers, or has any condition affecting speech production, LOGOS applies a condition modifier to the LCF — not to the NOSS sector activations. The NOSS output reflects what the oscillatory signatures say. The LCF accounts for the channel through which those signatures are expressed. These are different interventions at different layers.

Baselines established under pharmacological or physiological conditions that affect central processing (opioids, sedatives, stimulants) must be flagged as CONDITIONAL and require re-establishment under clean conditions for VERIFIED status.

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P-08.3 **Neurodevelopmental, gender, and disability calibration**

NOUS calibrates against the individual's own cognitive architecture, not population norms, for neurodevelopmental conditions. A subject with Down Syndrome or intellectual disability showing their stable, normal profile should receive S6 Optimal sector reading reflecting their healthy baseline — not a false-alarm pathological flag. Clinical value is in detecting *change from their baseline*, not in comparing them to neurotypical norms.

Gender identity is not a diagnostic finding. LOGOS should accept gender identity as an intake parameter to inform prosodic baseline construction — not to produce any clinical classification. A transgender woman's vocal baseline must be calibrated to her current vocal profile, not to sex-assigned-at-birth population data.

09 *Differential Signature Engine*

ARCHITECTURE

The Differential Signature Engine must distinguish between five etiological templates. These are not mutually exclusive — co-occurrence is clinically meaningful — but the primary etiology must be identified and distinguished. The distinguishing features below are the critical differential markers that the engine must implement:

ETIOLOGY	PRIMARY NOSS PATTERN	DISTINGUISHING FEATURE	PROSODIC SIGNATURE
Cerebrovascular	S4 elevation, executive-predominant	Decline WITHOUT prosodic-autonomic changes. Lateralized if focal.	Absent or minimal prosodic change
Cardiac	S4 elevation — but prosodic changes PRECEDE cognitive changes	Prosodic-autonomic changes appear 4–6 months before executive cognitive symptoms. Pathway 3 signature.	Pitch variability ↓, speech rhythm disrupted, HRV-correlated patterns
Neurodegenerative	S4+S8, motor-speech coupling	4.7 Hz focal tremor for Parkinson's. REM-associated dream narrative markers. Progressive trajectory.	Tremor frequency, reduced articulatory precision, hypophonia
Psychiatric	S1–S3, S9–S12 sector-specific	NOSS activation without trajectory degradation. Stable architecture with episodic fluctuation.	Varies by condition; prosody reflects emotional state not motor decline
Oncology / CRCI	S4 elevation, cyclic pattern	Chemotherapy: cycle-synchronized perturbation (pathognomonic). irAE: progressive NOT cycle-synchronized, 6–12 week onset.	Absent prosodic-autonomic signature (distinguishes from cardiac)

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Normal Aging	Global, slow, diffuse	No domain-specific predominance. Slow trajectory. No acute changepoints.	Mild slowing across all prosodic parameters

HARD RULE — CARDIAC VS. CEREBROVASCULAR DISCRIMINATION

The most critical differential is cardiac vs. cerebrovascular. The discriminating feature is the prosodic-autonomic temporal sequence: cardiac pathology produces prosodic changes BEFORE cognitive symptoms; cerebrovascular pathology does NOT. The Differential Signature Engine must implement this temporal check, not just a static sector comparison.

10 *Clinical Domain Expansion*

FIVE DOMAINS

NOUS is not a mental health tool. NOUS is a clinical cognitive surveillance platform that detects the cognitive consequences of the four leading causes of death and disability worldwide. The architecture must be designed to support all five clinical domains, even if the first release focuses on mental health.

P-10.1

Domain 1 — Mental Health (Foundation)

Full 12-sector NOSS coverage for psychiatric conditions. This is the validated origin domain with beta testing data (125+ practitioners). Primary NOSS sectors: S1, S2, S3, S5, S9, S10, S11, S12. CI, CSAA, and VOCA pathways all apply.

P-10.2

Domain 2 — Cerebrovascular

Detection of subclinical cerebrovascular compromise through S4 executive-predominant trajectory. Primary clinical value: subclinical TIA detection, progressive small vessel disease, stroke risk stratification. Key scientific claim: NOST can detect the 48–72 hour post-TIA cognitive signature even when the patient presents as asymptomatic. Longitudinal trajectory essential — cross-sectional assessment insufficient.

P-10.3

Domain 3 — Cardiovascular (Five Pathways)

Five cardiac detection pathways, each with distinct NOSS signatures:

```
Pathway 1: Reduced cardiac output → cerebral perfusion deficit → diffuse S4
Pathway 2: AF microembolic burden → progressive executive-attentional S4
Pathway 3: Autonomic dysregulation → prosodic changes PRECEDE cognitive S4
[EARLIEST]
Pathway 4: Systemic inflammation → gradual S4 degradation
```

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```
Pathway 5: Shared atherosclerotic burden → cerebral + coronary co-detection
```

Pathway 3 is the most clinically valuable because prosodic changes precede cognitive symptoms by 4–6 months. The Prosodic-Autonomic Analyzer must treat prosody as a primary signal for cardiac-domain assessments, not supplementary.

P-10.4

Domain 4 — Oncology (Cancer Care Continuum)

Four-phase monitoring architecture across the cancer care continuum:

```
Phase 1: Pre-treatment baseline (cancer-driven neuroinflammation already active
in ~30% at diagnosis)
Phase 2: Active treatment (cycle-synchronized monitoring for chemo neurotoxicity)
Phase 3: Recovery (post-treatment trajectory tracking)
Phase 4: Survivorship (long-term cognitive monitoring for 18M+ US cancer
survivors)
```

The **pathognomonic chemotherapy signature** — periodic cognitive perturbation synchronized with treatment cycles — is unique to this domain. No other etiology produces cycle-synchronized cognitive perturbation. When this pattern appears, chemotherapy neurotoxicity is the primary candidate.

Immunotherapy irAE sentinel: progressive cognitive decline NOT synchronized with treatment cycles, onset 6–12 weeks post-immunotherapy initiation, rate ≥3 S4 points/week → RED ALERT, suspected autoimmune encephalitis, urgent neurology referral.

P-10.5

Domain 5 — GENESIS (Prenatal / Maternal-Fetal)

Maternal cognitive monitoring during pregnancy and the perinatal period. Five prenatal pathways. MFM (Maternal-Fetal Medicine) escalation protocol. Gestational cognitive baseline tracking. This domain is the most recent addition and has a 36-month development roadmap.

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11 *The Objectivity Separation Principle*

NON-NEGOTIABLE

ARCHITECTURAL ABSOLUTE — ZERO COUPLING

Practitioner clinical notes must have ZERO coupling to the computational assessment layer. Not reduced coupling. Not weighted coupling. Zero. REMIEL channels, R2X translation, Beverly Index scoring, NOSS sector activations, DreamWeaver parameters, Memoria Sonata parameters, Empyrean rendering — none of these shall be aware that practitioner notes exist. They are architecturally blind to clinician input by design.

This principle is the single most important differentiator of NOUS from every other clinical decision support tool. The moment practitioner input influences computational output, the foundational value proposition — objective diagnostic data independent of clinician bias — is destroyed.

The analogy is radiology: a radiologist reads an MRI. The scan shows what it shows. The referring physician's clinical notes inform the radiologist's interpretation of the scan — but they do not change the pixels on the scan. NOUS works the same way. The Beverly Index score is the scan. The clinician's notes are the clinical context. Both exist in the record. Neither contaminates the other.

P-11.1

One permitted form of practitioner input: AAE protocol selection

There is one narrow exception to the zero-coupling rule: a practitioner may inform which AAE protocol runs (e.g., "patient is non-verbal — use alternative intake method," "patient has intellectual disability — apply developmental norms"). This changes HOW the assessment is conducted, not WHAT the assessment finds. It is an AAE configuration parameter, not a scoring modifier. It is logged in the AAE Configuration Card's adjustments list.

CODE IMPLICATION →

```
Practitioner Clinical Notes section: stores in a separate data layer. Zero API coupling to
assessment engine. Rendered in amber/gold in the dashboard to visually distinguish it from
all computational layers (blue/purple). The architectural disclaimer must appear whenever
the notes section is opened.
```

12 *Publication & IP Governance*

STRATEGIC

There is a strict tiered publication policy that governs what can and cannot be published without IP risk. This policy governs all external communication including papers, preprints, conference talks, and investor materials that describe technical methods.

P-12.1

Safe to Publish — Tier 1 (Theoretical Framework)

Papers describing what Computational Psychodynamics is, why NOST works in principle, what NOSS predicts, and how the 12-sector taxonomy is organized. These establish the field without exposing implementation. Think of it as publishing Einstein's general relativity without publishing the GPS engineering specs.

```
SAFE: Theory · Taxonomy · Field definition · Cross-domain analogues
SAFE: NOSS sector descriptions · Clinical correspondence · Testable predictions
```

Never Publish — Tier 3 (Trade Secrets)

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P-12.2

Never Publish — Tier 3 (Trade Secrets)

These are the proprietary implementation details that constitute the defensible competitive moat. They are trade secrets. They must never appear in any publication, paper, preprint, or public-facing document.

VIOLATION EXAMPLE →

NEVER PUBLISH: REMIEL 8-channel extraction architecture · R2X translation mechanics · Beverly Index scoring algorithms · CSAA authenticity ratio calculations · VOCA adaptive baseline methodology · LOGOS calibration weighting coefficients · Any training data or model weights

P-12.3

Six Targeted Utility Patents (not three umbrella patents)

The IP portfolio requires six targeted utility patents, each protecting a specific invention. Consolidating into fewer umbrella patents creates single-point-of-failure risk. The six patent claims:

```
Patent 1: NOSS Classification Method (12-sector computational classification)
Patent 2: NOUS Orchestrated Assessment System (full AAE→Beverly pipeline)
Patent 3: CSAA Forensic Authentication Method
Patent 4: VOCA Continuous Monitoring Method
Patent 5: LOGOS Cross-Cultural Calibration Method
Patent 6: Beverly Index Composite Scoring Method
```

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13 *Cross-Domain Scientific Analogues*

PUBLICATIONS ONLY

These findings from physics and astrophysics provide structural support for NOST's theoretical framework in academic publications. They are analogies — structural parallels, not claims of shared physical mechanism. Each entry below specifies appropriate deployment context.

CERN · QUARKONIUM RESONANCE SURVIVAL · 2025

Bound-state resonances persist as detectable signatures within quark-gluon plasma at temperatures far exceeding their expected dissolution point.

> Supports NOST Proposition 3 (Signature Specificity): cognitive-linguistic signatures persist under conditions of psychological stress, deliberate concealment, and deception. The subject cannot suppress their oscillatory signature any more than quark-gluon plasma can dissolve a quarkonium resonance at elevated temperature.

Tier 1 Papers — Proposition 3 section

JWST COSMOS-WEB · DARK MATTER COSMIC WEB · JANUARY 2026

Dark matter's invisible gravitational architecture determines where visible matter forms across the cosmic web. Invisible mass distributions determine visible structure distributions.

> The single strongest external analogy for NOST's core claim: unobservable oscillatory states determine the structure of observable linguistic output — not the other way around. Just as you cannot understand galaxy distribution without mapping the dark matter web, you cannot understand cognitive presentation without mapping the oscillatory signature beneath it.

Tier 1 Papers — NOST Information Principle section | Investor Presentations

TELESCOPE ARRAY / AMATERASU PARTICLE · 2021–2026

A 244 EeV cosmic ray appeared to originate from the Local Void — empty space with no known source capable of producing it. The signal was real. The source was invisible to every conventional observation method. Subsequent analysis suggests trajectory deflection by intervening magnetic fields obscured the true origin.

> Two NOST parallels: (1) Signal from the void — the most diagnostically significant cognitive signals originate from unconscious oscillatory states invisible to the subject, the interviewer, and every existing clinical instrument. The right detector array reveals them. (2) The deflection problem — a deceptive subject's narrative appears to originate from authentic recall; intervening "magnetic fields" (defense mechanisms, narrative management) bend the apparent trajectory. CSAA traces the signal back through the deflecting medium to identify true origin.

Tier 1 Papers — NOST Information Principle section | Keynotes and Presentations

CERN · QUARK-HADRON PHASE TRANSITION · 2025

A continuous chiral symmetry crossover (smooth underlying change) produces a sharp, sudden phase transition in the observable system.

> Supports NOSS activation threshold design in Paper 3: continuous oscillatory changes produce nonlinear threshold crossings when processing configuration reaches a critical point. Justifies discrete sector activation thresholds emerging from continuous underlying dynamics. A person doesn't gradually become psychotic — there is a threshold beyond which the cognitive signature reorganizes.

Paper 3 — Activation Model section

TUM · DEUTERON FORMATION · DECEMBER 2025

Fragile nuclear particles form not during the collision but after the fireball cools, from the decay of ultra-short-lived high-energy resonance states.

> Supports Lense-Thirring assessment timing principle: the most diagnostically informative cognitive signatures emerge during the post-acute consolidation period, not during peak crisis. A narrative collected 48 hours post-event may be more reliable than one collected during acute distress. The coherent signature doesn't form during the crisis — it crystallizes during cooling.

Paper 2 — Lense-Thirring section

SCOPE RESTRICTION — CROSS-DOMAIN ANALOGUES

These analogies are structural parallels, not mechanistic claims. Neural oscillations and cosmic particles share no physical substrate. Every published use of these analogies must be accompanied by an explicit disclaimer stating they are structural analogies only. Claiming shared mechanism would invite peer review rejection. The strength of these analogies is epistemological and rhetorical — they demonstrate that NOST's inferential architecture (detecting hidden generative states from observable secondary effects) is a

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universal scientific strategy, not an invention of Computational Psychodynamics.

14 *Academic Paper Portfolio*

SIX PAPERS

Six papers constitute the current academic publication portfolio. Tier 1 establishes the scientific field. Tier 2 extends the framework into specific clinical domains, each reaching a different medical specialty and a different potential partnership market.

ID	Title	Tier	Target Journal
Paper 1	Computational Psychodynamics: A New Framework for Objective Cognitive Assessment	Tier 1	*Psychological Review*
Paper 2	Neural Oscillation Signature Theory: Theoretical Foundations	Tier 1	*Neuroscience & Biobehavioral Reviews*
Paper 3	The NOSS Classification System: A 12-Sector Taxonomy for Computational Cognitive Assessment	Tier 1	*Psychological Assessment*
Paper 4	Linguistic Surveillance for Subclinical Cerebrovascular Compromise: A NOST Framework	Tier 2	*Stroke / Lancet Neurology*
Paper 5	Cardiocerebral Coupling and Linguistic Surveillance: A NOST Framework for Early Detection of Cardiovascular Pathology	Tier 2	*Circulation / JACC*
Paper 6	Linguistic Surveillance in Oncology: A NOST Framework for Cancer-Related Cognitive Impairment Detection, Treatment Neurotoxicity Monitoring, and Survivorship Cognitive Tracking	Tier 2	*JCO / Cancer / Neuro-Oncology*

Note: The Dream Code (published book) also constitutes a citable work and should appear on the ORCID record. All Tier 1 papers require patent attorney review before preprint upload to confirm no content extends beyond filed patent claims.

15 *RTCA — Full Architecture & Governance*

NON-NEGOTIABLE

Real-Time Call Analysis (RTCA) is the component that transforms NOUS from an episodic assessment platform into a continuous surveillance system. It is the only component that operates without the subject's conscious participation in producing input. This distinction — passive detection versus active assessment — defines its architecture, its ethics, its legal posture, and its commercial significance. None of these dimensions are optional. All must be governed explicitly.

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P-15.1

RTCA is a Sentinel, Not an Assessor — The Fundamental Distinction

Every other NOUS pathway — CI, CSAA, VOCA — requires a subject to consciously produce input. A dream narrative. A structured interview. A voice sample deliberately recorded. RTCA removes that requirement. It listens to natural, unstructured, ambient speech as it occurs: a phone call, a therapy session, a crisis line, a sports sideline conversation, an interrogation room, a military debrief.

RTCA does not assess. It detects and routes. This distinction is scientifically precise and legally critical:

```
RTCA function: ambient audio → rolling feature extraction → threshold comparison
→ routing decision
// RTCA produces NO Beverly Index score
// RTCA produces NO NOSS activation vector for clinical use
// RTCA produces ONE output: [ROUTE | CONTINUE MONITORING]
```

When RTCA's rolling activation pattern crosses threshold 65, it triggers AAE to initiate a structured CI or CSAA assessment. The structured assessment — not RTCA — produces the clinically actionable output. RTCA is the motion sensor that activates the camera. It is not the camera.

```
CODE IMPLICATION →
No clinical report, Beverly Index score, or NOSS vector shall be attributed to RTCA. All
clinical outputs are attributed to the CI or CSAA assessment that RTCA triggered. This
distinction protects regulatory positioning (RTCA is monitoring infrastructure, not a
diagnostic device) and legal defensibility (routing to assessment is not assessment).
```

P-15.2

Streaming Feature Extraction — The 30/15 Window Architecture

RTCA operates on a rolling window model fundamentally different from the batch processing used by CI and CSAA. A complete CI assessment processes a full narrative as a single input object. RTCA processes a continuous audio stream as a sequence of overlapping windows.

```
Window duration: 30 seconds
Window overlap: 15 seconds (50% overlap)
Advance rate: 1 new window every 15 seconds
Latency target: < 2 seconds per window (real-time constraint)
// Each window produces a partial NOSS vector
// Partial vectors aggregate across windows toward threshold
// No single window can produce a routing decision alone
```

The 50% overlap is architecturally essential. It ensures that a clinically significant utterance that spans a window boundary is captured in full by at least one window. A 30-second window with no overlap would lose any meaningful speech that begins in the last 5 seconds of one window and ends in the first 5 seconds of the next.

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HARD RULE — MINIMUM ACCUMULATION BEFORE ROUTING

RTCA must accumulate a minimum of 4 consecutive windows (covering 90 seconds of speech) before a routing decision can be triggered. A single high-activation window is insufficient. Requiring accumulation prevents false-positive routing from isolated expressive utterances, emotional emphasis, or transient stress responses that would resolve without clinical intervention.

CODE IMPLICATION →

The REMIEL channels active in RTCA mode are a constrained subset optimized for real-time performance. Full 8-channel extraction is not feasible at <2s latency. The RTCA channel set prioritizes: prosodic features (fast), sentiment valence (fast), lexical density (fast), and incoherence markers (fast). Deep syntactic parsing and semantic embedding — computationally expensive — run on the triggered CI/CSAA assessment, not on RTCA windows.

P-15.3 Speaker Diarization — Non-Negotiable for Multi-Party Audio

RTCA frequently operates on multi-party audio: a crisis line call (operator + caller), a therapy session (clinician + patient), a sports sideline conversation (trainer + athlete), an interrogation (examiner + subject). Without speaker diarization, the NOSS activation vector cannot be attributed to a specific individual. An activation spike caused by the crisis line operator's speech would be indistinguishable from one caused by the caller's — and routing a clinical assessment toward the wrong person is not a minor error.

```
Diarization requirement: identify and separate ≥ 2 speakers in real time
Attribution rule: RTCA activations track the designated SUBJECT speaker only
Subject designation: set at session initialization by operator or system
configuration
Fallback: if diarization confidence < 0.70, flag window as UNATTRIBUTED
// UNATTRIBUTED windows do not contribute to routing threshold accumulation
```

HARD RULE — DIARIZATION FAILURE PROTOCOL

If diarization confidence falls below 0.70 for more than 3 consecutive windows, RTCA must suspend routing decisions and flag the session as DIARIZATION DEGRADED. An operator alert is generated. No routing decision may be triggered on a DIARIZATION DEGRADED session. This prevents system misattribution under noisy, overlapping, or contested audio conditions.

P-15.4 The Activation Threshold — Architecture and Calibration

The RTCA routing threshold of 65 is not an arbitrary constant. It represents the activation level at which the probability of a clinically significant finding — one that would produce a Beverly Index Clinical Action Priority of MODERATE or higher — exceeds 80% based on calibration data. It is a probability gate, not a binary switch.

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```
Threshold: 65 (on normalized 0–100 RTCA activation scale)
Basis: P(CAP ≥ MODERATE | RTCA_activation ≥ 65) > 0.80
Deployment-specific adjustment: ± 10 points based on deployment context
// Crisis line deployment: lower threshold (55) — cost of false negative is high
// Corporate wellness deployment: higher threshold (75) — cost of false positive is high
// Military/intelligence deployment: operator-configurable within authorized range
```

Threshold calibration is a deployment parameter, not a scientific constant. The scientific principle is that the threshold represents a probability gate. The specific value is set based on the deployment context's tolerance for false positives versus false negatives, which varies significantly across markets.

CODE IMPLICATION →

The threshold value of 65 must be exposed as a configurable parameter in the RTCA deployment configuration, not hardcoded. Hardcoding it would prevent appropriate calibration across deployment contexts and create a liability when a crisis line deployment misses a call that a lower threshold would have caught.

P-15.5

Consent Architecture — The Ethical Framework Governing RTCA Deployment

RTCA is the most ethically complex component of NOUS because it operates on ambient audio from subjects who may not know they are being analyzed. This is not a problem to be avoided — it is a design constraint to be governed. The ethical framework below is non-negotiable. Any RTCA deployment that does not satisfy these requirements is not an authorized NOUS deployment.

TIER 1 — FULL INFORMED CONSENT DEPLOYMENTS

> Clinical therapy settings, telehealth platforms, sports medicine programs, occupational health programs. The subject is informed that audio analysis is active, the purpose is disclosed, and consent is documented before the session begins. RTCA operates as an opt-in enhancement to the clinical relationship. This is the standard deployment model for all non-investigative, non-governmental use cases.

TIER 2 — DISCLOSED BUT NON-SPECIFIC CONSENT DEPLOYMENTS

> Crisis lines, emergency services, call centers. The caller is informed that calls may be recorded and analyzed for quality and safety purposes — standard practice that has established legal precedent. RTCA analysis is included within this disclosure framework. The subject knows analysis is occurring; the specific nature of NOUS analysis is not detailed. This is the model for crisis line and emergency service deployments.

TIER 3 — AUTHORIZED NON-DISCLOSURE DEPLOYMENTS

> Intelligence community, law enforcement interrogation, military counterintelligence. Authorized by governing legal authority (court order, national security authorization, chain of command authorization). RTCA operates without subject disclosure. This deployment tier is available exclusively through government contract vehicles with appropriate legal authority documentation. Beverly Index LLC does not sell Tier 3 deployment to non-governmental actors under any circumstances.

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HARD RULE — NO UNAUTHORIZED NON-DISCLOSURE DEPLOYMENT

RTCA must never be deployed in a non-disclosure context without documented legal authorization. This includes but is not limited to: employer monitoring of employees without disclosure, spousal or relationship monitoring, parental monitoring of adults, law enforcement use without legal authority, and any deployment where the subject has a reasonable expectation of privacy and no consent framework is in place. Violations of this rule are not product misuse — they are legal violations that Beverly Index LLC will actively pursue.

P-15.6

Data Retention — RTCA Audio and Activation Records

RTCA processes audio in real time. The retention policy for that audio and for the activation records it generates is a governing scientific, legal, and ethical constraint — not an implementation convenience decision.

```
Raw audio: NOT retained by default (processed and discarded per window)
Window activation data: retained for 30 days, then purged (unless routing
triggered)
Triggered session data: retained per standard NOUS clinical data retention policy
Diarization metadata: retained for 7 days (audit purposes only)
// Exception: Tier 3 government deployments may have contract-specific retention requirements
// that supersede these defaults, governed by the authorizing contract vehicle
```

CODE IMPLICATION →

The default architecture must process RTCA audio in a streaming pipeline that does not write raw audio to persistent storage. Audio flows in, features are extracted, the audio is discarded. Only the derived feature vectors and activation scores are retained. This is both a privacy protection and a data minimization principle required by HIPAA and GDPR compliance.

P-15.7

Commercial Significance — The Markets Only RTCA Can Reach

RTCA is not a supplementary feature. It is the component that unlocks market segments that none of CI, CSAA, or VOCA can access, because those markets require ambient, passive, real-time analysis of naturally occurring speech. The structured intake process of CI, CSAA, and VOCA is operationally incompatible with these contexts.

CRISIS LINES

988 Suicide & Crisis Lifeline handles 5M+ contacts annually. RTCA can flag escalating calls in real time, enabling operators to elevate response before the subject self-identifies as being in crisis. The window between a call starting and a suicide attempt is measurable in minutes. RTCA operates in that window.

MILITARY & INTELLIGENCE

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Debrief sessions, interrogation support, insider threat detection programs, returned POW/hostage cognitive assessment. RTCA with Tier 3 authorization enables real-time CSAA triggering during sensitive interviews without requiring structured intake. The subject speaks. RTCA listens. AAE engages when warranted.

SPORTS MEDICINE SIDELINE

A player returning to the huddle after a hit speaks naturally to teammates and coaches. RTCA monitors that ambient speech for post-concussive cognitive signature. The team physician is alerted before the player self-reports symptoms — which concussed players frequently do not, under competitive pressure. This is the use case the NFL cannot currently address with any existing tool.

INSURANCE & LEGAL FORENSIC

Recorded claims interviews, legal depositions, recorded witness statements. RTCA operates on existing recordings (not real-time) to flag segments warranting CSAA forensic analysis. Investigators are directed to specific timestamps — not required to listen to hours of audio — where cognitive signatures suggest constructed versus authentic narrative. This is investigative triage at scale.

TELEHEALTH & REMOTE CARE

RTCA monitors telehealth video sessions continuously, alerting the clinician when the patient's speech during the session crosses clinical thresholds — even if the patient has not raised a concern. The clinician continues the conversation while receiving a private alert. This transforms the telehealth session from a conversation into a monitored clinical encounter.

CORPORATE & OCCUPATIONAL

High-stakes occupational contexts: nuclear facility operators, air traffic controllers, commercial pilots, financial traders in positions of fiduciary responsibility. RTCA in Tier 1 (disclosed) configuration provides continuous cognitive fitness monitoring during active duty shifts. An AEGIS integration routes flagged sessions to fitness-for-duty assessment automatically.

STRATEGIC IMPERATIVE

CI, CSAA, and VOCA are the assessment instruments. RTCA is the distribution infrastructure that delivers those instruments into markets where structured intake is operationally impossible. Without RTCA, NOUS serves markets where subjects can be directed to sit down and answer questions. With RTCA, NOUS serves markets where the subject is simply speaking — and NOUS is listening. That is a categorically larger addressable market.

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P-15.8

RTCA + AEGIS Integration — Continuous Fitness Monitoring

The most architecturally significant integration is RTCA + AEGIS for continuous cognitive fitness certification in occupational contexts. The standard AEGIS certification is point-in-time: a subject completes an assessment and receives CERTIFIED (8h), PROVISIONAL, or DENIED status. RTCA enables continuous certification — the subject's fitness is re-evaluated in real time throughout their duty period based on naturally occurring speech.

```
RTCA_session_start: subject begins duty shift
```

```
RTCA_continuous: monitors ambient speech throughout shift
RTCA_activation ≥ threshold: triggers AEGIS re-evaluation
AEGIS_result = DENIED: generates supervisor alert + duty suspension
recommendation
// Continuous AEGIS via RTCA is distinct from point-in-time AEGIS via CI
// Both are valid certification pathways; context determines which applies
```

For nuclear facility operators, air traffic controllers, and commercial pilots — where a single cognitive impairment event during a duty shift can produce catastrophic consequences — continuous AEGIS via RTCA represents a qualitatively different safety architecture than any pre-shift screening protocol. Pre-shift screening certifies the person at the start. Continuous monitoring certifies the person throughout.

RTCA ARCHITECTURE SUMMARY

CLASSIFICATION	Deployment Pathway Not an intelligence module
FUNCTION	Sentinel & Router Detects, does not assess
WINDOW MODEL	30s / 15s overlap 4 windows minimum before routing
ROUTING THRESHOLD	65 (default) Configurable 55–75 by deployment
DIARIZATION	Required ≥ 0.70 confidence Suspends on degradation
AUDIO RETENTION	Not retained (default) Features only, streaming discard
CONSENT TIERS	3 tiers (Full / Disclosed / Authorized) Tier 3 requires legal authorization
AEGIS INTEGRATION	Continuous fitness certification Occupational safety contexts
EXCLUSIVE MARKETS	Crisis lines · Military · Sideline Forensic · Telehealth · Occupational

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NOUS Platform — Scientific Principles Document v1.1
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23 Trademarks · 35+ Patent Applications

"We built the first system that can hear disease in the way people speak and see pathology in the way people dream. We're not replacing clinicians — we're giving them a cognitive stethoscope."

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